MICAH KEEGAN

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EDUCATION

University of California, Berkeley – College of Engineering & College of Letters and Sciences May 2029
B.S. Electrical and Computer Engineering · B.A. Physics

Relevant Coursework: Physics E&M + Mechanics/Special Relativity, Computational Techniques in Physics, Program Structures, Data Structures, Machine Structures, Calc I/II/III, Lin Alg/Diff Eq, Signals & Linear Systems

PROFESSIONAL EXPERIENCE

EECS Course Staff Berkeley, CA
Teaching Intern - CS 61C (Machine Structures) *August 2026 – Present*

- Lab tutor & discussion TA providing individual support to 30+ students weekly, reinforcing core concepts in C programming, assembly, parallelism, RISC-V microarchitecture, caching, pipelining, & number representation
- Enrolled in supplementary pedagogy course; developed & refined concept mapping of 61C to improve lecture pacing

Ferrari Berkeley, CA
Contract Backend Engineer *January 2026 – May 2026*

- Engineered data pipeline using Scrapy/Playwright/BeautifulSoup, normalizing 5,000+ manufacturer records from 8+ databases into PostgreSQL schema on AWS RDS; paired w/ FAISS semantic vector index over supplier profiles
- Automated outreach via Twilio SMS + SendGrid email APIs to contact 200+ suppliers & constructed Plotly Dash

PEAR Volunteering Santa Clara, CA
Technical Lead, Head of Outreach *July 2024 – August 2025*

- Architected & shipped full-stack platform on Next.js/React + PostgreSQL (Prisma) w/ Google OAuth for tokenized third-party verification; built RESTful API routes: opportunity geolocation, hour tracking & verification, email alerts
- Conducted stakeholder interviews w/ administrators & students to scope MVP; led 2025-26 pilot w/ 22 seniors each meeting graduation requirements; 2026-27 launch to serve 1,700+ students at net-zero cost, with goals to scale further

49ers STEM Leadership Institute Santa Clara, CA
Student Leadership Committee President *Dec 2024 – May 2025*

- Nominated & elected president of internal student-directed team; directly mentored in leadership by Chevron executives
- Organized 12 leaders to develop & teach curriculum to 240 academically inclined students across 6 grades; implemented mentorship program increasing cross-cohort engagement by 300%; led fundraisers resulting in $760,000+ raised for SLI

PROJECTS & RESEARCH

FPGA Probabilistic-Bit Ising Machine for Hardware-Accelerated Optimization August 2026 – Present

- 32-spin probabilistic-bit Ising machine in SystemVerilog for RealDigital AUP-ZU3: fixed-point stochastic p-bit units with piecewise-linear tanh approximation, BRAM-backed coupling matrix, & per-spin LFSR pseudorandom generators

2D Molecular Dynamics Simulation with Lennard-Jones Potential August 2025 – January 2026

- Numba-JIT-compiled physics kernels (∼500x speedup), cell-list neighbor search w/ linked-list traversal (O(N^2) to O(N) force compute), velocity-Verlet symplectic integration, & periodic boundary conditions via minimum-image convention
- Demonstrated convergence of particle speeds to theoretical Maxwell-Boltzmann distribution from non-equilibrium initial conditions; quantified accuracy via L1, chi-squared, & max-bin error metrics (<0.05 for 500,000 particles), benchmarked CPU vs projected GPU scaling, & built ImageIO-streamed GIF pipeline for memory-efficient trajectory visualization

Nuclear Spin Initial-State Optimization for NMR Power Absorption September 2024 – May 2025

- Experimental pipeline built in Python/Anaconda using PULSEE (Brown University quantum simulation library built using QuTiP), ran 108 trials across 27 initial-state configurations (3^3 coupled-spin combinations) $\times$ 4 pulse durations: constructed Hamiltonians (Zeeman, quadrupolar, EM coupling), applied resonant RF pulses via Liouville-von Neumann evolution, generated FID signals & extracted NMR spectra via Fourier transform for quantitative density matrix analysis

ADDITIONAL INFORMATION

Languages: C/C++, Python, SQL, JavaScript/HTML/CSS, Java, Rust, Verilog, SystemVerilog, LaTeX
Frameworks: CUDA, React/Next, Node.js, AWS (RDS), Prisma ORM, Flask, Django, Tailwind, hls4ml
Tools: Git, Docker, Conda, Vim, Chipyard, KiCAD, Solidworks, Fusion 360, COMSOL, Vivado
Libraries: Pandas, NumPy, Numba, Matplotlib, OpenCV, PyTorch, Sci-Kit, QuTiP, TensorFlow